Egyptian Academy for Engineering & Advanced Technology

Mechatronics Engineering Department

“Automated Liquid Mixing Plant (MixFlow)”

Course Name

SCADA and Automation Systems - MEC 483

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1. Abstract

MixFlow is an automated liquid mixing plant designed and simulated using the Ignition SCADA platform. The system consists of three liquid supply tanks, a mixing tank, and a finished product tank. Each supply tank is equipped with a pump and a control valve to regulate the transfer of liquids into the mixing tank.

The system monitors tank levels and controls the mixing process through a sequence of operations, including liquid filling, mixing, and product transfer. The mixing tank has a total capacity of 300%, allowing it to receive the required quantities of the three liquids. After the mixing process is completed, the final product is transferred to the finished product tank.

The project aims to demonstrate the application of SCADA technology in industrial process monitoring and control through a simulated environment. It provides a visual representation of the system, enabling operators to monitor tank levels, control pumps and valves, and observe the different stages of the process.

Project in github.com

Keywords

PID Controller, Collision Avoidance System, Distance Control, Ultrasonic Sensor, DC Motor Control, Autonomous Vehicles, Servo Motor Control, Robotic Arm, Firefighting Robot, Water Pump System, Arduino Uno, MATLAB Control Interface

2. Introduction

Industrial automation plays an important role in improving the efficiency, accuracy, and reliability of industrial processes. Supervisory Control and Data Acquisition (SCADA) systems are widely used to monitor and control industrial operations by providing a graphical interface that allows operators to observe process conditions and manage system components.

The MixFlow project focuses on the design and simulation of an automated liquid mixing plant using the Ignition SCADA platform. The system consists of three liquid supply tanks, a mixing tank, and a finished product tank. Each supply tank is equipped with a pump and a control valve to regulate the flow of liquid into the mixing tank.

The mixing process is designed to operate through a sequence of stages, including filling the mixing tank with the required liquids, mixing the contents, and transferring the final product to the finished product tank. The system monitors the tank levels and controls the operation of pumps and valves according to the selected process conditions.

The main objective of this project is to demonstrate the use of SCADA technology in industrial process monitoring and control through a simulated environment. The project provides a graphical interface for monitoring tank levels, controlling system components, and observing the different stages of the liquid mixing process. This simulation helps illustrate the basic principles of process automation and the role of SCADA systems in managing industrial operations.

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3. Objectives

- To design and simulate an automated liquid mixing plant using the Ignition SCADA platform.
- To monitor the liquid levels in the supply tanks, mixing tank, and finished product tank.
- To control the operation of pumps and valves for regulating the transfer of liquids between tanks.
- To implement a sequential mixing process that includes filling, mixing, and transferring the finished product.
- To provide a graphical Human-Machine Interface (HMI) for monitoring and controlling the system.
- To demonstrate the application of SCADA technology in industrial process automation through simulation.
- To improve the understanding of process monitoring, control logic, and automated operation in industrial systems.

4. Software Tools and Technologies



1. Ignition SCADA Platform

Ignition was used to develop the SCADA application and create the graphical interface for monitoring and controlling the liquid mixing plant. It provides the environment for designing the HMI and implementing the system logic.

2. Ignition Vision Module

The Vision module was used to design the operator interface, including the graphical representation of tanks, pumps, valves, and other system components. It allows the operator to interact with the simulated process through buttons and visual indicators.

3. Tags and Expressions

Tags were used to represent the process variables, such as tank levels, pump states, valve states, and mixing status. Expression tags and expressions were used to implement the control logic and simulate the behavior of the system components.

4. Human-Machine Interface (HMI)

The HMI provides a visual interface through which the operator can monitor the liquid levels, control the pumps and valves, and observe the different stages of the mixing process.

5. System Implementation and Simulation Results

5.1 Before Operation

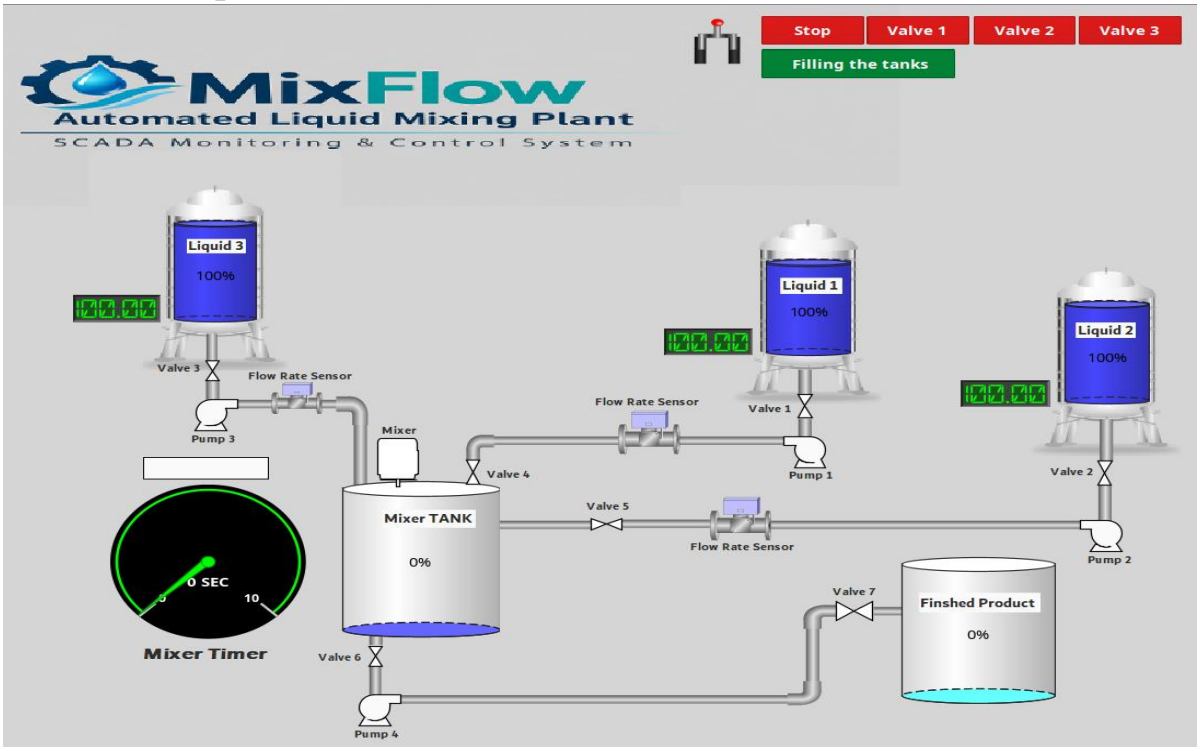


Figure 1: MixFlow SCADA system before starting the liquid mixing

5.2 During Operation

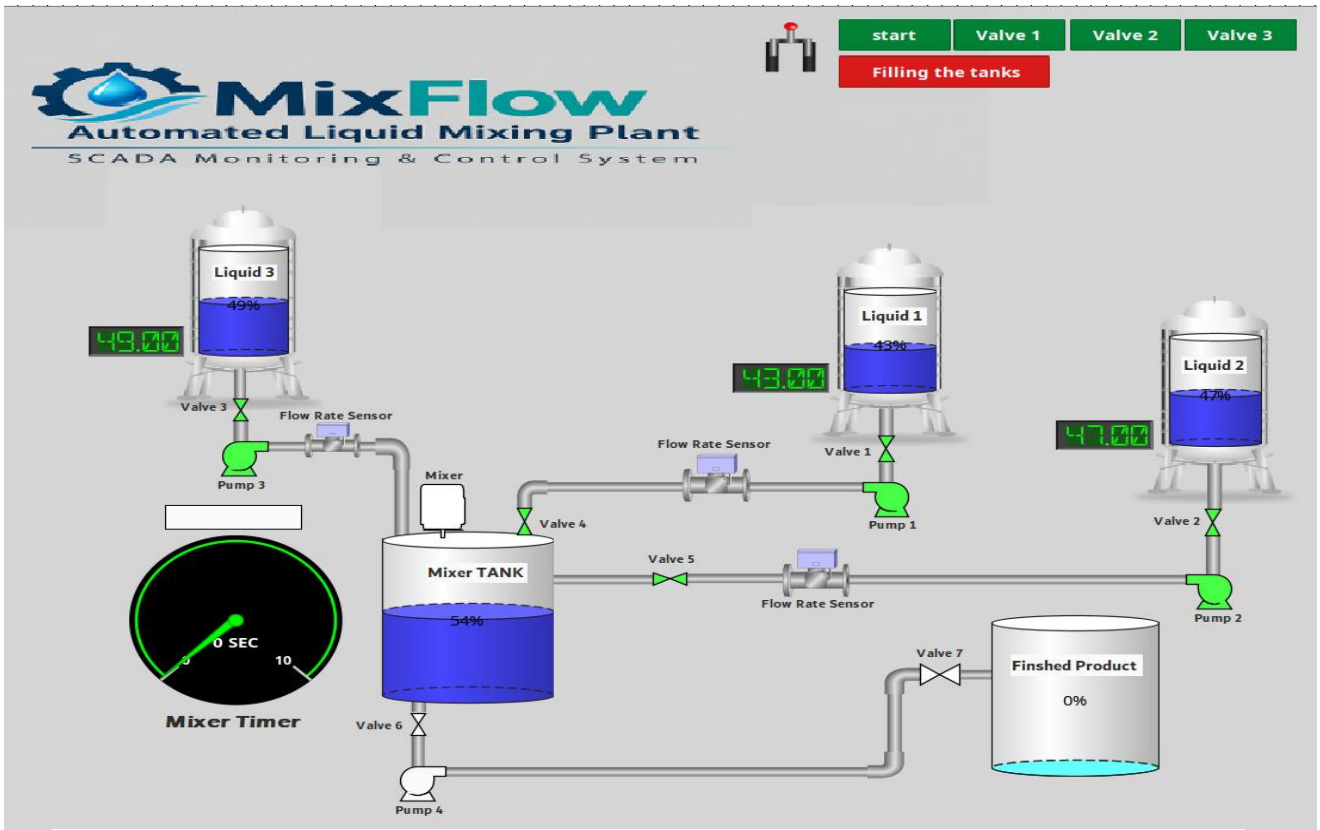


Figure 2: MixFlow SCADA system during the liquid filling and mixing process.

5.3 After Operation

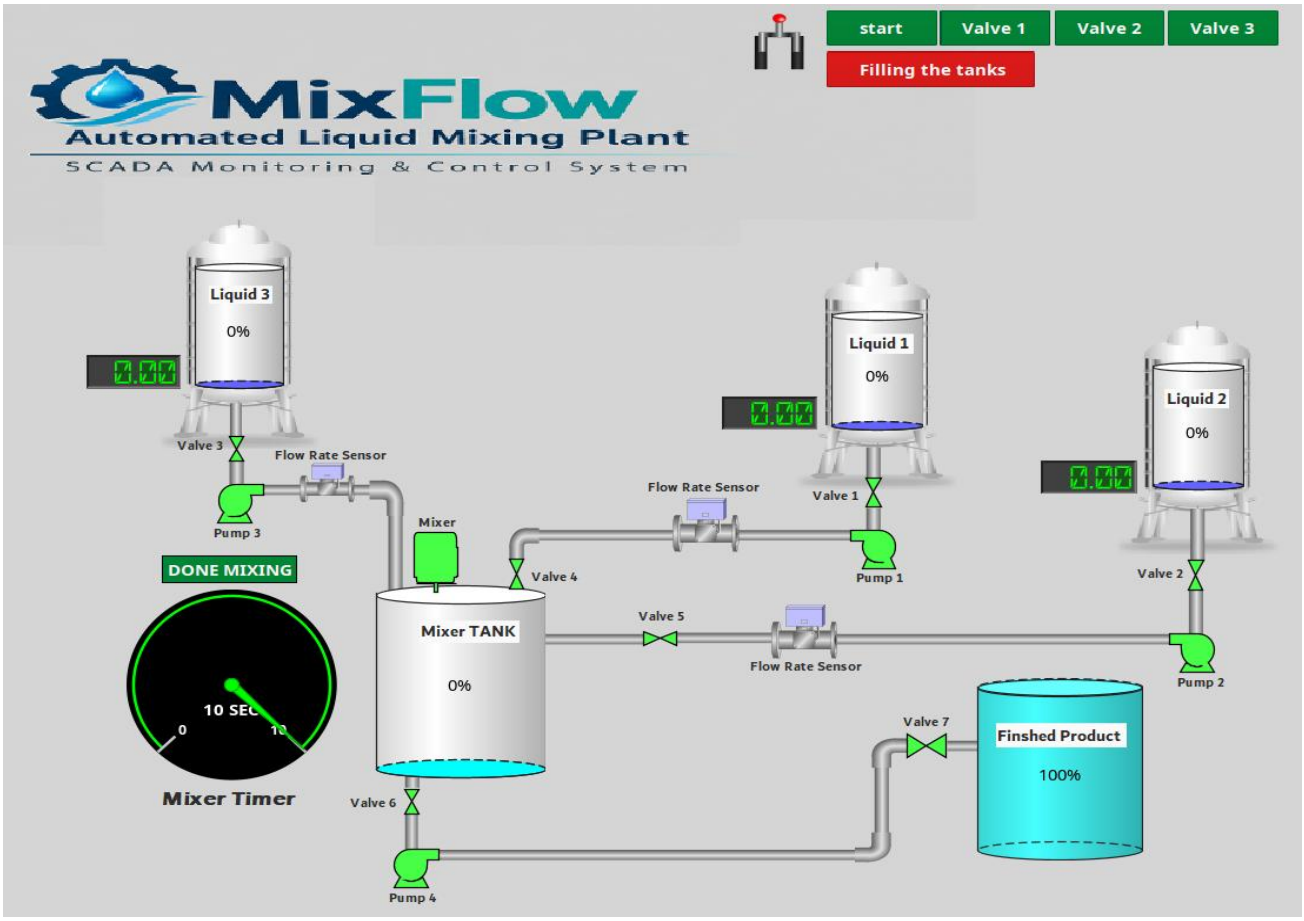


Figure 3: MixFlow SCADA system after completing the mixing and product transfer process.

6. Tags Description

Tags are used in the MixFlow SCADA system to represent and monitor the different process variables. They also provide the values required to control the operation of the tanks, pumps, valves, and mixing process.
The following table describes the main tags used in the system.

Tag Name	Description
TANK1	Represents the liquid level in the first supply tank.
TANK2	Represents the liquid level in the second supply tank.
TANK3	Represents the liquid level in the third supply tank.
TANK4	Represents the liquid level in the mixing tank.
TANK5	Represents the liquid level in the finished product tank.
V1	Controls the valve connected to the first liquid tank.
V2	Controls the valve connected to the second liquid tank.
V3	Controls the valve connected to the third liquid tank.
start	Represents the start command for the filling process.
Filling	Indicates the filling or refill state of the system.
MixerStart	Indicates the start of the mixing process.
MIXERTIMER	Represents the mixing process counter.
MIXINGDONE	Indicates that the mixing process has been completed.

The tag values are used by the system expressions to simulate the behavior of the process. For example, when a control valve is opened and the filling process is active, the corresponding supply tank level decreases while the mixing tank level increases.

The tags also allow the system to control the sequence of operations, including filling the mixing tank, mixing the liquids, and transferring the finished product to the final tank.

Tag Editor

TANK1

Expression Editor: TANK1 > Expression

```
if(isNull({[~]TANK1}),100,if({[]Filling}=1 && {[]start}=0 && {[]V1}=0 && {[~]TANK1}<100,
min(100,{[~]TANK1}+1),
if({[]start}=1 && {[]V1}=1 && {[~]TANK1}>0,{[~]TANK1}-1,{[~]TANK1})))
```

Tag Editor

TANK2

Expression Editor: TANK2 > Expression

```
if(isNull({[~]TANK2}),100,if({[]Filling}=1 && {[]start}=0 && {[]V2}=0 && {[~]TANK2}<100,
min(100,{[~]TANK2}+1),
if({[]start}=1 && {[]V2}=1 && {[~]TANK2}>0,{[~]TANK2}-1,{[~]TANK2})))
```

Tag Editor

TANK3

Expression Editor: TANK3 > Expression

```
if(isNull({[~]TANK3}),100,if({[]Filling}=1 && {[]start}=0 && {[]V3}=0 && {[~]TANK3}<100,
min(100,{[~]TANK3}+1),
if({[]start}=1 && {[]V3}=1 && {[~]TANK3}>0,{[~]TANK3}-1,{[~]TANK3})))
```

Tag Editor

TANK4

Expression Editor: TANK4 > Expression

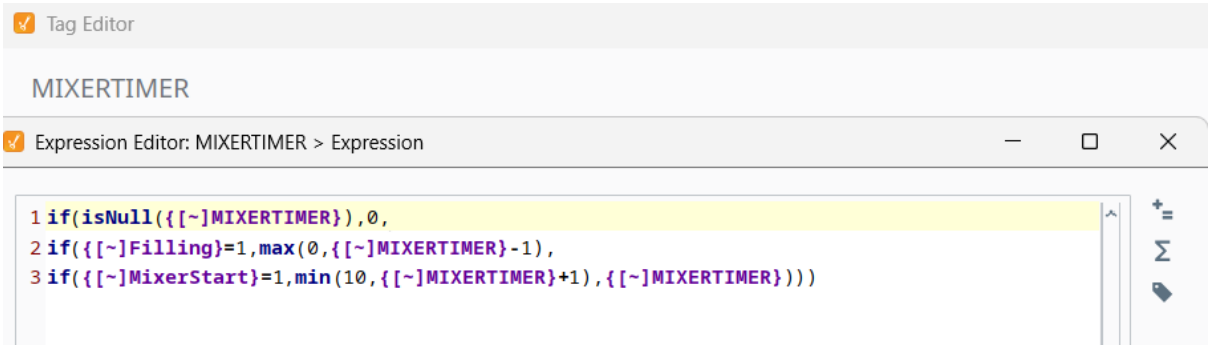
```
if(isNull({[~]TANK4}),0,if({[]MIXINGDONE}=1 && {[~]TANK4}>0,max(0,{[~]TANK4}-1),
if({[]Filling}=1 && {[~]TANK4}>0,max(0,{[~]TANK4}-1),
if({[]start}=1,min(300,{[~]TANK4}+
if({[]V1}=1 && {[~]TANK1}>0,1,0)+
if({[]V2}=1 && {[~]TANK2}>0,1,0)+
if({[]V3}=1 && {[~]TANK3}>0,1,0)),{[~]TANK4}))))
```

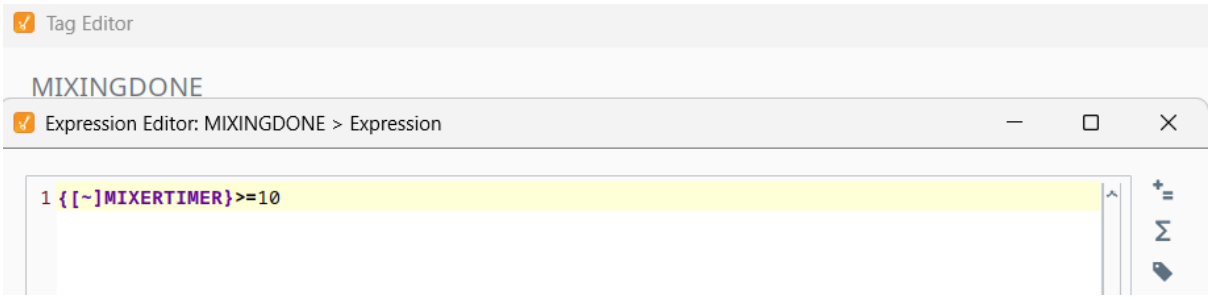
Tag Editor

TANK5

Expression Editor: TANK5 > Expression

```
if(isNull({[~]TANK5}),0,if({[~]Filling}=1,max(0,{[~]TANK5}-1),
if({[~]Filling}=0 && {[~]MIXINGDONE}=1,
min(300,{[~]TANK5}+1),{[~]TANK5})))
```


Tag Editor
MIXERTIMER
Expression Editor: MIXERTIMER > Expression
1 if(isNull({[~]MIXERTIMER}),0,
2 if({[~]Filling}=1,max(0,{[~]MIXERTIMER}-1),
3 if({[~]MixerStart}=1,min(10,{[~]MIXERTIMER}+1),{[~]MIXERTIMER})))


Tag Editor
MIXINGDONE
Expression Editor: MIXINGDONE > Expression
1 {[~]MIXERTIMER}>=10

7. Conclusion

The MixFlow project demonstrates the application of SCADA technology in monitoring and controlling an automated liquid mixing plant using the Ignition platform. The system was designed to simulate the operation of three liquid supply tanks, a mixing tank, and a finished product tank.

Through the use of tags, expressions, and a graphical HMI, the project simulates the main stages of the process, including liquid filling, mixing, and transferring the final product. The system also provides visual monitoring of tank levels and control of pumps and valves throughout the operation.

The project helped demonstrate the role of SCADA systems in industrial process automation and provided practical experience in designing a simulated process control system. Overall, MixFlow provides a clear example of how industrial processes can be monitored and controlled through a user-friendly SCADA interface.

8. References

1. Inductive Automation, *Ignition User Manual – Vision Module*. Available at: https://www.docs.inductiveautomation.com/docs/8.1/ignition-modules/vision
2. Inductive Automation, *Ignition User Manual – Types of Tags*. Available at: https://www.docs.inductiveautomation.com/docs/8.3/platform/tags/types-of-tags
3. Inductive Automation, *Ignition User Manual – Tag Bindings in Vision*. Available at: https://www.docs.inductiveautomation.com/docs/8.1/ignition-modules/vision/binding-types-in-vision/tag-bindings-in-vision
4. Inductive Automation, *Ignition User Manual – Expression Binding in Vision*. Available at: https://www.docs.inductiveautomation.com/docs/8.1/ignition-modules/vision/binding-types-in-vision/expression-binding-in-vision
5. Inductive Automation, *Inductive University – Tags in Ignition*. Available at: https://inductiveuniversity.com/videos/transcript/tags-in-ignition
6. Inductive Automation, *Inductive University – Tag Binding*. Available at: https://inductiveuniversity.com/video/tag-binding/8.1
7. Inductive Automation, *Ignition Documentation and Tutorials*. Available at: https://inductiveuniversity.com/
8. Inductive Automation, *Ignition Official YouTube Channel*. Available at: https://www.youtube.com/@InductiveAutomation